TUNISIAN REUBLIC
Ministry of Higher Education and Scientific Research
University of Carthage
National Institute of Applied Sciences and Technology

End of Year Project

Second Year Engineering Cycle

Specialty: Computer Networks and Telecommunications

Intelligent Multimodal approach for Automatic Generation of High-Impact Short Formats (Shorts / Reels)

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Academin Year: 2025/2026

Acknowledgments

Abstract

Write your abstract here.

Keywords: ..., ...

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- **Abbrev.** Abbreviation

Introduction

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State of the Art

Introduction

Video understanding is an inherently multimodal problem. A highlight is rarely defined by a single signal it emerges from the alignment, or sudden divergence, of what is seen, heard, and spoken simultaneously. Building a system capable of detecting such moments therefore requires models that can reason jointly across modalities, not process them in isolation. This chapter traces the progression from unimodal approaches and their limitations, through fusion-based systems, to joint multimodal representation learning the paradigm this work builds upon. Each step in this progression is motivated by the shortcomings of the one preceding it, converging on the architectural foundation of the proposed approach.

1 Unimodal Approaches for Video Understanding

Video understanding has traditionally been approached through unimodal analysis, where only a single modalityvisual, audio, or textis processed independently. While such approaches can extract meaningful information, they often fail to capture the full semantic and emotional richness of long-form videos.

For highlight detection and segmentation tasks, relying on a single modality may lead to incomplete or misleading interpretations of what constitutes a high-impact segment.

1.1 Visual Understanding

Visual-based methods focus on analyzing video frames to extract spatial and temporal patterns. Early systems relied on handcrafted features, while modern approaches use deep convolutional neural networks to learn visual representations automatically.

Typical visual analysis techniques include:

- Frame-level feature extraction

- Optical flow for motion estimation

- Shot boundary detection

- Scene classification

- Object and action recognition

Libraries such as OpenCV and Decord are widely used for frame extraction and temporal sampling in video processing pipelines. However, visual saliency does not necessarily correspond to semantic importance. A visually dynamic segment may lack narrative relevance, while a static scene containing crucial speech content may represent a key moment. Therefore, visual-only segmentation is insufficient for detecting meaningful high-impact clips.

1.2 Audio Understanding

Audio-based approaches analyze the acoustic signal to detect speech, music, sound events, and emotional cues. These systems typically rely on spectral and temporal representations

of sound.

Common techniques include:

- Mel-Frequency Cepstral Coefficients (MFCCs)

- Spectrogram analysis

- Energy-based segmentation

- Pitch and prosodic feature extraction

- Speaker diarization

The Python library Librosa is commonly used for extracting spectral and rhythmic features such as MFCCs, chroma features, spectral centroid, and tempo descriptors.

Although audio intensity can signal emotionally charged moments, it may also produce false positives due to background music or noise. Furthermore, impactful segments may not always correspond to peaks in acoustic energy. Hence, audio-only systems lack contextual grounding.

1.3 Textual Understanding

Text-based video understanding relies on Automatic Speech Recognition (ASR) to convert spoken content into transcripts, enabling Natural Language Processing (NLP) analysis.

Modern ASR models such as Whisper provide robust transcription for long-form and multilingual audio content.

Once transcribed, NLP techniques can be applied for:

- Keyword extraction

- Topic segmentation

- Sentiment analysis

- Emotion detection

- Semantic similarity analysis

Textual analysis enables identification of semantically important moments in videos. However, it ignores visual expressiveness, cinematic effects, and acoustic intensity, which

are often crucial in determining whether a segment is engaging or impactful.

2 Representative Unimodal Models and Frameworks

2.1 Whisper

2.2 Librosa

2.3 OpenCV / Decord

3 Limitations of Unimodal Approaches for Highlight detection

For the task of segmenting long videos into short, high-impact reels, unimodal systems face significant limitations:

- Visual systems detect motion but not meaning.

- Audio systems detect intensity but not narrative relevance.

- Text systems detect semantics but not emotional amplification.

High-impact segments typically emerge from the interaction between modalitiesfor example:

- A powerful statement (text) delivered with emotional tone (audio)

- A dramatic visual scene reinforced by intense background music

- A key announcement accompanied by audience reactions

Therefore, unimodal analysis fails to model cross-modal dependencies that define impactful video moments.

4 Fusion-Based Multimodal Systems

5 Joint Multimodal Representation Learning

Joint multimodal representation learning addresses a fundamental limitation shared by both unimodal and fusion-based approaches: the absence of a unified semantic space in which modalities such as vision, audio, and text can be collectively treated and compared. Rather than processing each signal independently and combining outputs,the joint approach trains a single model to vimap all modalities into a shared high-dimensional embedding space, where geometric proximity reflects semantic similarity regardless of the originating modality.

Modern multimodal systems are inspired by the contrastive learning framework popularized by CLIP-like architectures, enabling cross-modal reasoning where heterogeneous modalities are treated as different expressions of the same underlying objective. Joint multimodal representation learning addresses the core challenge of cross-modal alignment, enabling direct comparison and retrieval across modalities without relying on engineered fusion rules or modality-specific supervision.

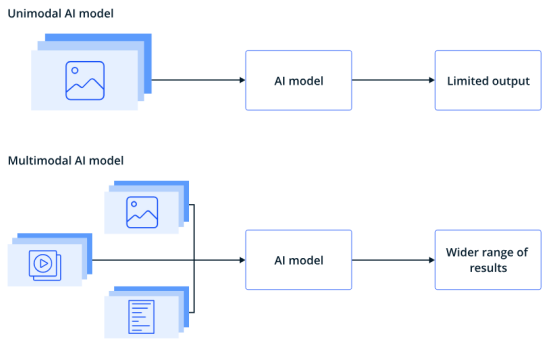

Figure 1.1: Comparison of unimodal and multimodal processing frameworks. Unimodal systems process a single data type, while multimodal systems integrate multiple modalities into a unified representation [AI Planet, 2024].

5.1 Architectural Overview

Although joint multimodal models vary in design, most share a common three-stage architecture: modality-specific encoders, a fusion mechanism, and optionally a decoder for generative tasks Sajid [2024].

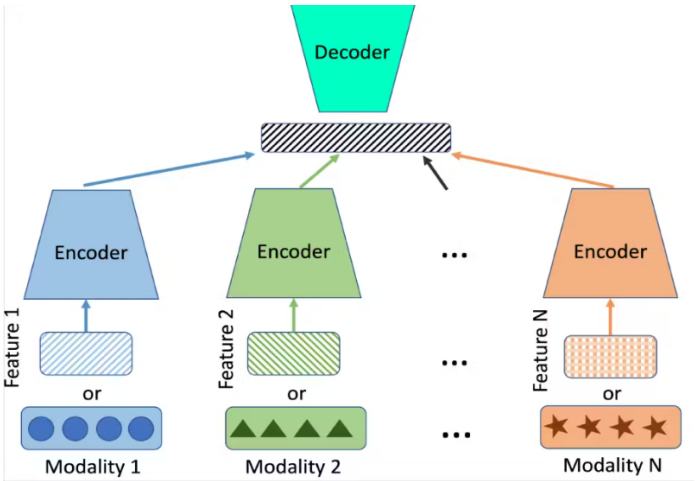

Figure 1.2: Joint multimodal model Architecture
Sajid [2024]

Modality-specific encoders transform raw inputs into dense feature vectors: image encoders employ **CNNs** or **Vision Transformers (ViT)**, text encoders leverage **transformer architectures** to map token sequences into contextual embeddings, and audio encoders such as Wav2Vec2 [Baevski et al., 2020] convert raw waveforms into representations capturing rhythm, tone, and semantic content. Once encoded, a fusion mechanism integrates these embeddings into a unified representation. Four strategies are commonly used: *early fusion* combines raw features before processing; *intermediate fusion* merges latent representations; *late fusion* combines final modality outputs; and *hybrid fusion* applies elements of all three at different stages. These strategies are implemented through attention-based methods Vaswani et al. [2017], feature concatenation, or dot-product interactions between modality vectors.

5.2 Contrastive Learning Framework

Contrastive learning is the fundamental training strategy underlying modern multimodal models. Rather than predicting explicit class labels, the model learns by comparing pairs of inputs: a **positive pair** consists of semantically aligned samples from different modalities — an image and its corresponding caption, or a video frame and its paired audio segment — while a **negative pair** consists of mismatched samples. The model is trained to pull positive pairs closer together in the embedding space while pushing negative pairs apart, with the supervision signal arising entirely from cross-modal correspondence rather than manual annotation.

This is formalized through the InfoNCE objective [van den Oord et al., 2019]. Given a batch of N matched pairs $(x_i^{(A)}, x_i^{(B)})$ from two modalities A and B, each encoder maps its input to a **normalized** embedding such that the correct match is identifiable among all N candidates. The symmetric loss is:

$$\mathcal{L} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{e^{\mathrm{sim}(\mathbf{e}_i^A,\mathbf{e}_i^B)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^A,\mathbf{e}_j^B)/\tau}} + \log\frac{e^{\mathrm{sim}(\mathbf{e}_i^B,\mathbf{e}_i^A)/\tau}}{\sum_j e^{\mathrm{sim}(\mathbf{e}_i^B,\mathbf{e}_j^A)/\tau}}\right] \tag{1.1}$$

where $\mathrm{sim}(\cdot,\cdot)$ is cosine similarity and τ is a temperature hyperparameter controlling the sharpness of the similarity distribution. A lower τ sharpens the objective, producing a more discriminative but noise-sensitive embedding space; in practice, τ is either fixed — CLIP uses $\tau = 0.07$ [Radford et al., 2021] — or learned per modality, as in ImageBind [Girdhar et al., 2023]. This framework has proven highly scalable, enabling multimodal pretraining on large-scale web data without any manually labelled supervision.

5.3 Shared Embedding Spaces

The product of contrastive training is a shared embedding space, typically of dimension $d \in \{512, 768, 1024\}$, in which each modality is processed by a dedicated encoder — a transformer-based language encoder for text, a vision transformer for images, or a spectrogram-based encoder for audio [Sajid, 2024] — followed by a projection layer that maps the resulting features into the same latent dimensionality. All embeddings are constrained to lie on the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 **normalization**, making cosine similarity a universal proximity measure across modalities:

$$\mathrm{sim}(\mathbf{e}^A, \mathbf{e}^B) = \mathbf{e}^A \cdot \mathbf{e}^B \qquad \text{since } \|\mathbf{e}\| = 1 \tag{1.2}$$

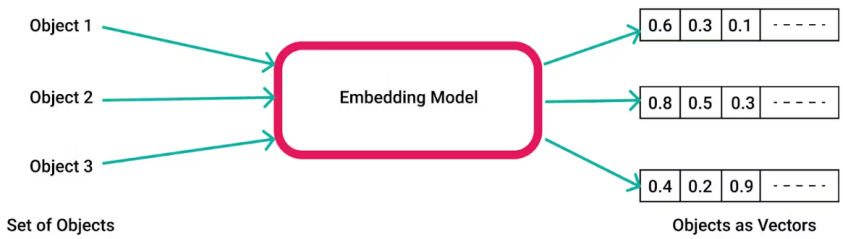

Figure 1.3: Multimodal embeddings: raw inputs from different modalitiesprojected into a common high-dimensional vector space Sajid [2024].

Within this shared space, semantically related inputs are positioned close to one an-

other regardless of modality: an image of a cheering crowd, an audio clip of applause, and a textual description of excitement occupy nearby regions of the embedding manifold. This unified representation facilitates cross-modal similarity computation and retrieval without requiring explicit modality-specific fusion mechanisms.

A further property is *emergent transitivity*: a model trained on image-audio and image-text pairs separately achieves implicit audio-text alignment through the shared visual anchor, without the two modalities ever being trained together directly.
ImageBind [Girdhar et al., 2023] exploits this property to bind six modalities — image, text, audio, depth, thermal, and IMU — using only image-paired data for each, achieving full cross-modal alignment across all modality combinations at inference time.

5.4 Zero-Shot Transfer and Cross-Modal Retrieval

The shared embedding space directly enables zero-shot transfer: the ability to perform recognition or scoring on novel concepts expressed as natural language, without any task-specific training. Because the model learns semantic alignment rather than task-specific boundaries, it can transfer to unseen categories without fine-tuning. A concept such as "crowd applause" is encoded by the text encoder into a semantic anchor embedding; candidate segments are then scored by their cosine similarity to this anchor, regardless of modality:

$$\text{score}(x) = \text{sim}(f_{\text{text}}(q),\ f_{\text{target}}(x)) \tag{1.3}$$

The shared space enables cross-modal retrieval: a natural language query can retrieve the most semantically relevant video segment purely on embedding proximity, while an audio clip of crowd cheering can retrieve the visual moment of a goal.
This "search by anything" capability supports text-to-image, image-to-text, audio-to-text, and audio-to-video retrieval within a single framework.

5.5 Advantages and Limitations of Joint Multimodal Learning

Joint multimodal learning paradigm offers four key advantages over alternative approaches.It provides **native cross-modal alignment**, to encode semantic correspondences directly in the geometry of the embedding space rather than through engineered fusion rules. It enables **zero-shot generalization**, allowing the system to handle new

content categories at inference time through natural language, without depending on annotated datasets. It supports **modality-agnostic scoring**, where a single similarity function operates uniformly across all available modalities.

Finally, models (such as ImageBind) benefit from **large-scale pretraining** on internet-scale paired data, encoding world knowledge that transfers effectively to downstream tasks without fine-tuning.

Despite these strengths, the paradigm presents notable limitations. The most significant is the **absence of temporal modeling**: joint multimodal models produce a single embedding per fixed-length segment and cannot model sequential dynamics across segments. Narrative structures, such as the comedic buildup across several exchanges before a punchline, must be handled by external temporal analysis. A second limitation is **embedding norm uniformity**: because all embeddings are ℓ_2-normalized, their magnitude carries no information, making raw energy-based scoring inapplicable without additional statistical processing. Finally, **modality imbalance** in pretraining data: image-text pairs, being far more abundant than audio-centric combinations, can reduce the discriminative power of audio embeddings in content where acoustic cues are the primary signal.

Addressing these challenges remains an active area of research, particularly for video-centric applications where temporal coherence and multimodal interaction play a central role.

6 Representative Multimodal Models

6.1 CLIP

6.2 CLAP

Contrastive Language-Audio Pretraining (CLAP), as described in [Elizalde et al., 2022], is an extension of the contrastive learning paradigm of CLIP to the audio domain, with the aim of learning a joint representation space between audio and text to facilitate zero-shot audio classification and retrieval tasks. The architecture employs two encoders to process the audio and text modalities in the form of mel spectrogram and text, respectively, and

is trained using symmetric contrastive loss to maximize the similarity between matching and minimize the similarity between non-matching audio and text pairs.

Despite its success in aligning the audio and text modalities, the limitations of the CLAP approach for the task of highlight detection in videos are as follows: the audio is processed in isolation without access to the visual information, thus making it impossible to leverage the audio-visual modalities. In addition, the encoders are optimized for short videos, thus limiting the modeling of long-range dependencies, which is critical in the case of videos. The use of architectures such as ImageBind, which combines the three modalities in a single representation space, can be more effective for multimodal video analysis.

6.3 Flamingo

The Flamingo model developed by DeepMind is a powerful family of Visual Language Models designed to process sequences of images or videos interleaved with text and generate free-form text as output.
Flamingo represents a notable progression in vision-language modeling by enabling a single model to perform multiple multimodal tasks with minimal task-specific fine-tuning.

A core concept of its architecture is the fusion of large, pre-trained, and frozen models specifically a **vision encoder** and a **language model**to leverage their existing knowledge without the need for intensive retraining .
To efficiently bridge these two modalities, Flamingo introduces **the Perceiver Resampler**, which compresses a huge number of visual features into a fixed-size collection of visual tokens , these tokens then condition the frozen language model through **Gated Cross-Attention layers**, which are interleaved between the original language model layers , these gated layers are essential for preserving the stability of the model during training and ensuring the language model's original performance remains intact at initialization.

A key contribution of Flamingo is its **few-shot learning** capability The model can adapt to various tasks with only a few input-output examples . Experimental evaluations showed strong performance across several visionlanguage benchmarks, such as multimodal reasoning tasks, visual question answering and image captioning .

6.4 LLAVA

LLaVA is a multimodal model designed to enhance general-purpose visual and language understanding through the use of visual instruction tuning

The architecture of LLaVA consists of a **pretrained CLIP vision encoder** and a **Vicuna large language model**, which are connected by a two-layer **Multi-Layer Perceptron cross-modal projector** .
This MLP projector converts visual encoder tokens into post-embedding layer LLM tokens, replacing the simpler linear projection used in earlier versions to improve multimodal capabilities .
The training process follows a **two-stage pipeline:** an initial stage focused on modality alignment using roughly 600,000 image-text pairs, followed by a second stage of visual instruction fine-tuning where the model is trained on diverse conversation datasets .
This approach enables LLaVA to perform tasks such as image description, visual question answering, and multimodal dialogue generation. Compared to more complex architectures, LLaVA is considered more accessible and computationally efficient while maintaining competitive performance.

6.5 Video-LLaMA

Video-LLaMA [Zhang et al., 2023] is a video understanding model that uses the generative method of incorporating video content into large language models. Unlike embedding-based video retrieval models that treat text, vision, and audio as equivalent modalities in an embedding space, Video-LLaMA places the language model at the core of the model and treats vision and audio as conditionals for text generation. The model is instruction-tuned to process video content with synchronized audio for natural language reasoning on dynamic video content.

The model employs two branches: the Vision-Language branch that aggregates video features using Video Q-Former to extract temporal dependencies across frames using cross-frame attention mechanisms, and the Audio-Language branch that processes audio content using ImageBind [Girdhar et al., 2023] to extract features that can be used in conjunction with the language model.
Although Video-LLaMA is effective in processing temporal video content for scene un-

derstanding and question answering tasks, its autoregressive generation method is not appropriate for embedding-based video retrieval tasks. However, the training of video language models like Video-LLaMA is computationally expensive and demands large-scale video-text paired datasets, which limits its adoption by many research groups.

VideoLLaMA 2 [Cheng et al., 2024] is an improved version of VideoLLaMA that uses a spatial-temporal convolution connector instead of the dual branches of the original model to link the vision and audio models to the language model and improves audio content processing using BEATs [Chen et al., 2022].

6.6 ImageBind

7 Comparative Analysis and Discussion

8 Positioning of the Proposed Approach

The trajectory surveyed in this chapter points to a single conclusion: meaningful cross-modal reasoning requires modalities to be aligned at training time, not combined. The approach adopted in this work follows directly from that constraint.

8.1 Paradigm Justification

Unimodal approaches, while effective within their domain, fail to capture the semantic richness of video, which is inherently multimodal. Fusion-based methods combine representations learned independently, which limits their ability to model deep cross-modal correspondences and to generalize to scenarios unseen during training.

Joint multimodal representation learning is the paradigm that supports direct tri-modal comparison without modality-specific supervision. By projecting vision, audio, and text into a shared embedding space through contrastive training, it becomes possible to measure cosine similarity across heterogeneous modalities in a consistent manner.

This mechanism is precisely what underpins the scoring strategy adopted in our pipeline: **a segment is considered salient when it exhibits strong joint activity or a significant semantic shift across consecutive modalities**.

8.2 Model Justification

Among the joint embedding models surveyed, ImageBind 6.6 stands out as the most suitable choice for the problem at hand. CLIP 6.1 is limited to the image-text pair and has no audio processing capability. CLAP 6.2 covers audio and text but disregards the visual modality. Video-LLaMA 6.5, while integrating all three modalities, relies on a large language model backbone that makes it ill-suited for dense segment-level inference.

ImageBind provides a unified embedding space trained around vision as the anchor modality, enabling alignment across six modalities without pairwise supervision. For our use case trimodal embedding extraction over two-second segments at the scale of a full video this model offers the optimal trade-off between modality coverage, computational compactness, and a well-documented API.

8.3 Identified Limitations and Methodological Handling

Two limitations of ImageBind inform the pipeline design. Its lack of temporal attention time being encoded solely through 3D convolutional patches reduces sensitivity to intra-segment dynamics. Silent segments further risk producing uninformative audio embeddings that distort the final score. The adressing of these limitations is detailed in Chapter 2.

Conclusion

2

Design and Implementation

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4 Missing Modality Handling: Text Imagination

5 Highlight Scoring and Selection

6 Evaluation Metrics and Validation Protocol

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Appendix 1

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